

Hishaam Ali Abbasi

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PROFESSIONAL SUMMARY

Robotics engineering graduate with hands-on experience in mechanical design, embedded systems, firmware development, and control systems. Proven ability to take projects from concept to working prototype, from a high-fidelity medical training device to a production home server running containerised services. Strong foundation in CAD, 3D printing, Python, C++, and Linux infrastructure. Seeking a graduate engineering role.

TECHNICAL SKILLS

Programming: Python (Advanced), Java (Advanced), C++ (Intermediate), SQL, C, HTML/CSS/JavaScript

Embedded & Control: ESP32, Arduino, Raspberry Pi, ROS2, PID control, kinematics, haptics, NVENC

CAD & Manufacturing: Onshape, SolidWorks, Creo, Inventor, Fusion 360, Ansys FEA, 3D printing (FDM/Orca), AutoCAD

Infrastructure: Linux (Debian/Ubuntu), Proxmox VE, Docker, LXC, Nginx, Cloudflare Tunnel, AWS S3/Glacier, systemd

Software & Tools: Git, MS Office, Adobe Photoshop/Illustrator, DaVinci Resolve, Portainer, Nextcloud, rclone

Languages: Native English, Conversational Urdu

EXPERIENCE

Engineering Design and Research Intern — Alvis Robotics, Birmingham Aug 2025

- Debugged firmware and diagnosed mechanical failure points on bomb disposal (EOD) robot platforms.
- Supported R&D, legacy system overhauls, and financial management alongside the Managing Director.

HSBC Spring Insight Programme — Quantitative Stream Jul–Aug 2024

- Analysed financial datasets and explored quantitative modelling workflows over a 3-week programme.

Clinical Data Administrator — AT Medics, London Jul–Aug 2021

- Processed large clinical datasets and improved data governance, eliminating operational backlogs.

COVID Clinic Front of House — AT Medics, London Dec 2020–Mar 2021

- Registered patients and assisted non-English speakers through the vaccination process.

PROJECTS

Home Server and Self-Hosted Infrastructure 2025–Present

- Built a Proxmox VE hypervisor on a Xeon workstation with GPU passthrough (Nvidia RTX 3050) for hardware transcoding, running Docker containers via LXC for Jellyfin, Nextcloud (4 users, 8TB), Nginx Proxy Manager, and automated AWS S3 Glacier backups.
- Configured Cloudflare Tunnel and Zero Trust Access across 5 domains; migrated 4 websites from Vercel to self-hosted Nginx; resolved kernel/driver, AppArmor, and cgroup passthrough challenges.

Fracture Immobilisation Trainer — Dissertation 2025–2026

- Engineered a fluidic pump architecture with subcutaneous force sensors and ESP32-based real-time pain index for medical trainee education. Combined rapid prototyping, embedded systems, and haptics.

Formula Student — Suspension Team, Queen Mary Racing Oct 2023–Present

- Optimised suspension geometry using Ansys FEA; contributed to full redesign for a new car platform.

Other Projects

Active camera gimbal (C++ PID control), robotic arm, BoxTurtle multi-material 3D printing system, web development, e-commerce shop. Full portfolio and case studies at [hishaamabbasi.co.uk](#).

EDUCATION

BEng Robotics Engineering — Queen Mary, University of London 2022–2026

Key modules: Numerical Methods and Data Science (86%), Control Systems and Analysis (78.9%), Modelling and Control of Robotic Systems (75.2%), Design for Human Interaction (70.2%).

Sutton Grammar School 2015–2022

A-Levels: Mathematics (A), Physics (A), Computer Science (B). 12 GCSEs incl. Maths (9), CS (8), Physics (8).

LEADERSHIP AND ADDITIONAL

- Explorer Scout Leader (2012–present): Chief Scout Diamond Award. Founded new Explorer section. Plan and run weekly sessions, expeditions, and camps.
- Combined Cadet Force Flight Sergeant (2018–2023): Taught syllabus, led teams, mentored cadets.
- Full UK driving licence.